Mohamed Sucule

Frontend Engineer · React · TypeScript · Accessibility

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PROFILE

Frontend engineer with 4+ years' commercial experience shipping web applications for product and consultancy teams, including SC-cleared public-sector programmes at Capgemini. I build in React and TypeScript, treat accessibility and performance as requirements rather than polish, and test what I ship: the two apps below each have strict types, unit and end-to-end tests, and their own CI.

EXPERIENCE

Software Engineer · Capgemini — July 2022 – Present

- Build and maintain production web applications and services in TypeScript, JavaScript and Python, connected to back ends through REST APIs, cloud infrastructure and CI/CD pipelines.
- Turn business and user requirements into technical solutions with senior stakeholders and delivery teams.
- Work across the whole delivery lifecycle: technical discovery, implementation, testing, deployment and live support.
- Deliver technology for major public- and private-sector clients, including programmes under SC security clearance.

Web Developer · SynchrTrade — 2021 – 2022

- Built and refined responsive front-end features in React and JavaScript within agile sprints, working closely with backend engineers and designers.
- Balanced usability, implementation constraints and delivery speed across a commercial product lifecycle.

SELECTED PROJECTS

Cells · a spreadsheet written from scratch — Live

React 19, TypeScript (strict), Vite, Vitest, fast-check, Playwright

- Wrote the formula engine (a Pratt parser and 116 Excel-compatible functions) and a dependency graph that recalculates only what changed.
- Profiled a 200,001-cell recalculation from 681 ms to 133 ms by moving the graph work onto typed arrays.
- Virtualised ARIA grid of 1,048,576 rows with Excel's keyboard and touch editing; 209 unit tests, including a property test against a reference evaluator, and 16 end-to-end tests.

Your Place · any place on Earth as a print-ready map poster — Live

React 19, TypeScript (strict), Vite, Web Workers, OffscreenCanvas, Vitest, Playwright

- Wrote the vector-tile decoder and renderer from scratch; it paints in a Web Worker, so the interface never stutters.
- Profiled and fixed two bottlenecks, taking the first poster from 41.7 s to 0.39 s; exports 300 dpi PNG and SVG from one geometry pass.
- Accessible place search (WAI-ARIA combobox); 62 unit tests, 12 end-to-end scenarios and axe checks.

Portfolio · 37 projects that run in the browser — Live

JavaScript, Canvas, WebGL, Web Audio, GitHub Actions

- Engineering tools (a regex engine you can watch execute, an OKLCH palette builder with contrast checks, a Raft consensus simulator), games and in-browser machine-learning apps.
- Over 1,000 tests run in CI before every deploy, with automated accessibility checks on every page.

SKILLS

Frontend	React, TypeScript, JavaScript (ES2022+), HTML, CSS, responsive design, accessibility (WCAG 2.1 AA, WAI-ARIA), Canvas 2D, SVG, WebGL, Web Workers, Web Audio
Testing & tooling	Vitest, Playwright, axe-core, ESLint, Prettier, Vite, Git, GitHub Actions, GitLab CI/CD
Back end & cloud	REST APIs, Python, FastAPI, AWS, Azure
Delivery	Agile delivery, technical discovery, stakeholder management

CERTIFICATIONS

AWS Certified Cloud Practitioner · Microsoft Azure AI Fundamentals (AI-900) · GitLab Certified

EDUCATION

BEng Mechanical Engineering · Aston University, Birmingham — 2021